Question-conditioned writing in image space

Imagized Language Model Project

Methods companion, 5 November 2026. Question-dependent writing is a requirement for an independent image-language model. This note specifies a registered paired training objective and its completed implementation preflight. Autonomous-language performance is not established by these implementation checks.

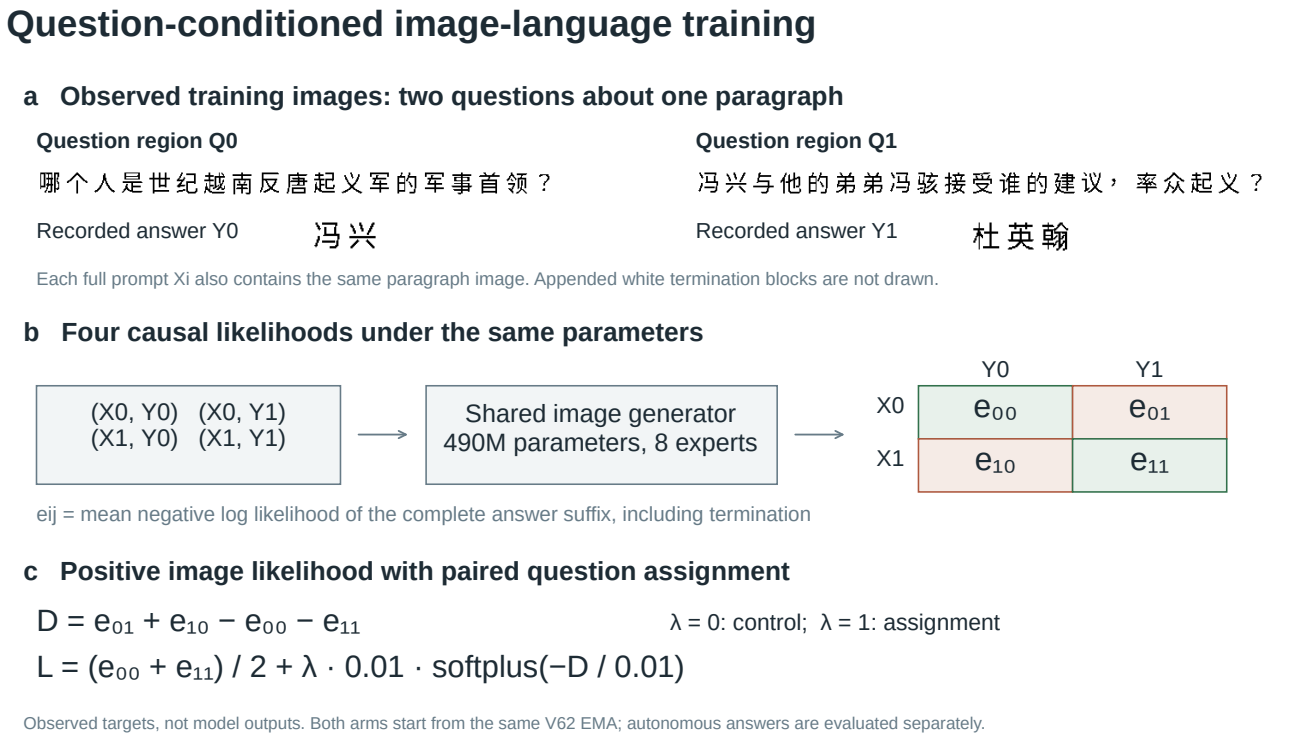


Fig. 1 | Paired image-language training. a, Question regions and observed answer images from the first admitted training pair. Each full prompt also contains the shared paragraph, omitted from display. Targets retain native padding; appended white termination blocks are not drawn. These are training data, not model-generated answers. **b,** One image generator scores all four causal prompt–answer streams. Diagonal cells are own-question targets; off-diagonal cells are swapped targets. The e_{ij} labels are symbolic energies, not numerical measurements. **c,** Positive image likelihood is supplemented by an assignment loss. The two arms share initialization, architecture, sampled pairs and update budget, differing in λ. Actual free answers and question controls are evaluated separately.

Physical observations and targets

The source is a short-answer Chinese reading subset of CMRC 2018 [1]: 3,211 training questions and 1,321 reused development questions, with original article overlap removed. Training pairs share a paragraph and have disjoint retained answer strings and canonical reference-image sequences. Admission retains 3,793 pairs across 939 paragraphs and 3,016 distinct questions. Two pairs sharing an accepted answer string are excluded. Paragraphs, then valid pairs within them, are sampled uniformly.

Each X_i is the complete observed prompt image. Each Y_j is a padded answer-image suffix followed by two white termination blocks. The actor reads binary 32×64 physical image blocks; it receives no character IDs, answer location, glyph inventory or external language-model response. Host rendering of source text is data preparation, outside the actor boundary.

Conditional image probability

The contextual backbone is externally pretrained MIXAR-477M [2], adapted with eight conditional bitmap experts. All 489,998,208 actor parameters participate in training. For an answer block with 2,048 white/ink bits (one denotes white),

$$p_\theta(y_t \mid h_t) = \sum_{k=1}^{8} \pi_k(h_t) \prod_{u=1}^{2048} \mathrm{Bern}\big(y_{tu}; \sigma(a_{ku}(h_t))\big). \qquad (1)$$

The gate is normalized, and every expert defines a normalized bitmap law. Thus the mixture is normalized. Each expert is a context-dependent neural function, with no fixed character identity. The causal state h_t sees the prompt and preceding answer blocks. Future target blocks cannot enter an earlier prediction. The suffix probability is the product of these block conditionals.

Paired objective

Let n_j be the number of scored pixels in the complete suffix Y_j, including termination. Define

$$e_{ij} = -\frac{1}{n_j} \log p_\theta(Y_j \mid X_i), \qquad (2)$$

$$D = e_{01} + e_{10} - e_{00} - e_{11}, \qquad (3)$$

$$L = \frac{e_{00} + e_{11}}{2} + \lambda\, \tau\, \mathrm{softplus}(-D/\tau), \quad \tau = 0.01. \qquad (4)$$

The control uses $\lambda = 0$; assignment training uses $\lambda = 1$. Both arms compute the same four likelihood graphs. Teacher forcing

supplies observed answer pixels only after their causal predecessors.

Adding arbitrary row and column offsets $a_i + b_j$ to e_{ij} leaves D unchanged, because each offset occurs once with each sign. The derivative of the assignment term with respect to D is $-\sigma(-D/\tau)$. This elementary identity describes a training signal, not a theorem of semantic learning. For equal $n_0 = n_1 = n$, D equals $1/n$ times the log ratio of correct to swapped joint assignments. With unequal lengths, it is a length-normalized interaction energy and must not be called that single joint log odds or a mutual-information estimate. Feature interactions such as question/answer width can still support ranking without useful writing.

Registered comparison

Both arms start from the same learned V62 eight-expert EMA. Each makes 2,000 updates, accumulating four pairs per update: eight positive and eight swapped streams. AdamW uses learning rate 2×10^{-5}, betas $(0.9, 0.95)$, weight decay 0.01, 100-step warmup, cosine decay to one tenth, norm clipping at one and EMA coefficient 0.99. Parameters, optimizer and EMA use FP32; forward/backward use BF16 autocast, deterministic eager attention and nonreentrant activation checkpointing. The final EMA is fixed in advance; development scores do not select checkpoints or seeds. Arms run sequentially on physical GPU 0, with a 20-GiB allocated-memory and eight-hour limit per arm. External foundation pretraining is separate from local adaptation cost.

Completed implementation preflight

All 26 implementation checks pass using training data only. A full optimizer and EMA update on the longest admitted joined stream, 250 image blocks with all four graphs retained, peaks at 9.197 GiB. Four-update checks peak at 9.235 GiB for either arm. Four uninterrupted updates exactly match two plus resumed two in model, EMA, optimizer, Python/NumPy/Torch CPU/CUDA RNG, pair RNG and the complete sampled-pair journal, independently for each arm.

All parameter gradients are finite. Assignment-only gradients are nonzero through the physical input projection, first/last contextual layers, bitmap projection and mixture gate. Physical four-stream assembly matches the source prefix/suffix tensors. Changing future answer pixels gives zero earlier-output change. Strict-FP32 cached/prefill maximum logit difference is 6.485×10^{-5}, within the registered 2×10^{-4} tolerance. These checks establish implementation feasibility; they do not establish language accuracy or efficiency at matched capability.

Inference and independent assessment

The deployed actor generates physical bitmap candidates, scores them under the complete mixture and feeds selected pixels back into its visual context. Searching expert threshold modes is approximate MAP, not a guaranteed global mode. An all-white block stops writing. No malformed image is repaired, snapped to a character bank or replaced by a font glyph.

The locked audit follows both completed training arms and independent offline deployment replay. It retains 256 primary questions, all 393 shared-paragraph pairs and all 1,321 development likelihood cases, with normal and blank-question controls. The bounded reading requirements include at least 10% exact primary answers, 25% strict-raster F1, 2% both-correct pairs, fewer than 20% capped outputs and a positive lower bound of the 95% article-bootstrap interval for the question effect. The assignment intervention additionally requires a positive lower bound for its raster-F1 gain over control. Reference-answer energy ranking is diagnostic and cannot replace autonomous answers. No outcome from that audit is reported in this methods companion.

Canonical font shift, reused development, one training seed, externally pretrained weights and short extractive answers limit interpretation. Useful general Chinese/English instruction following, books and historical writing remain distinct requirements. Code, fixed protocol, preflight and image-source hashes accompany this note; the complete primary record remains in the repository.

References

1. Cui, Y. et al. A Span-Extraction Dataset for Chinese Machine Reading Comprehension. *Proceedings of EMNLP-IJCNLP*, 5883–5889 (2019). `https://aclanthology.org/D19-1600/`.
2. Hu, C., Tai, Y., Vergari, A., Keller, F. & Suglia, A. MIXAR: Scaling Autoregressive Pixel-based Language Models to Multiple Languages and Scripts. *arXiv* 2604.11575v2 (2026). `https://arxiv.org/abs/2604.11575v2`.